



Jonathan Gonzalez

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EDUCATION

Bachelor of Science: Computer & Data Science (GPA 3.96)

Dec, 2025

University of Indianapolis

Indianapolis, IN

Relevant courses: Big Data Mining, Applied Statistical Methods, Intro to Neural Networks, Database Systems, Statistical Learning, Bayesian Analysis and Statistical Decision Making

High School Diploma with Honors

May 2022

West Lafayette Jr/Sr High School

West Lafayette, IN

SKILLS

Git - Python (Pandas, scikit-learn) - SQL - Deep Learning - Tableau - AWS - Data Analysis - Statistics - Bilingual

WORK EXPERIENCE

Student Researcher

May 2024 - Dec 2025

University of Indianapolis

Indianapolis, IN

- Developed and evaluated machine learning models, including linear models and CNNs, to simulate neuronal responses to auditory stimuli
- Achieved a model performance of $R^2 = 0.57$, demonstrating statistically meaningful predictive capability on experimental data
- Performed data cleaning, preprocessing, and feature engineering on time-series sensory datasets
- Conducted model validation and error analysis to assess generalization and reliability

Mathematics & Engineering Tutor

Nov 2022 - Dec 2025

University of Indianapolis

Indianapolis, IN

- Providing individual and group tutoring sessions to students in a variety of math courses
- Assisted students with C++, Python, and problem-solving techniques
- Assessing student understanding and adapting teaching methods to meet individual student needs

Roche Academy Intern (Software Developer)

May 2025 - Aug 2025

Roche

Indianapolis, IN

- Designed and implemented an AI-driven system to support engineers with instrument troubleshooting using structured and unstructured data
- Built end-to-end data pipelines to ingest raw communication logs, preprocess text, and generate structured knowledge artifacts
- Redesigned the help-article creation workflow, increasing productivity by 125× compared to previous systems, dramatically reducing time-to-knowledge delivery
- Developed a human-in-the-loop quality assurance application to validate AI-generated outputs before deployment
- Implemented an AI agent architecture integrating an LLM, task-specific tools, and decision logic to automate information retrieval

IT Assistant

May 2022 - Oct 2024

Exploradoor

Indianapolis, IN

- Supported data-driven digital tools and internal software systems
- Followed structured processes for system configuration, troubleshooting, and documentation
- Improved operational efficiency through reliable technical support and clear communication

CERTIFICATIONS

- International Tutor Training Program Certification: Level I
- Certified in TypeScript: Object Oriented Programming and Asynchrony
- Certified in OOP & Algorithms in Python

SCHOLARSHIPS AND AWARDS

- ILI full-tuition Scholarship
- REID Foundation Scholarship (Software Engineering)
- Emerging Mathematical Greyhound
- Dewey E. Gommel Award for Excellence in Mathematics

PROJECTS

Auditory Processing Analysis

GitHub: [Auditory_Processing](#) — Python, Data Modeling, Signal Processing

- Built a neural data processing and analysis pipeline for auditory stimuli using Python, preparing, cleaning, and ingesting neural response datasets.
- Implemented algorithms to compute firing rates and extract patterns, supporting exploratory and predictive modeling in neuroscience domains.

evGrandPrix Simulator - Software Lead

GitHub: [evGrandPrix-Simulator](#) — LISP, Python, Distributed Systems

- Served as primary software developer for the EV Grand Prix Simulator in a multidisciplinary project to emulate the physical dynamics and real-time behaviors of a go-kart racing system.
- Designed and implemented key simulation software components, including data ingestion, event loops, control logic, and vehicle state systems in LISP.
- Collaborated with hardware and design teams to ensure simulation accuracy aligned with physical parameters, emphasizing modular, extensible C++ code for embedded/real-time environments.